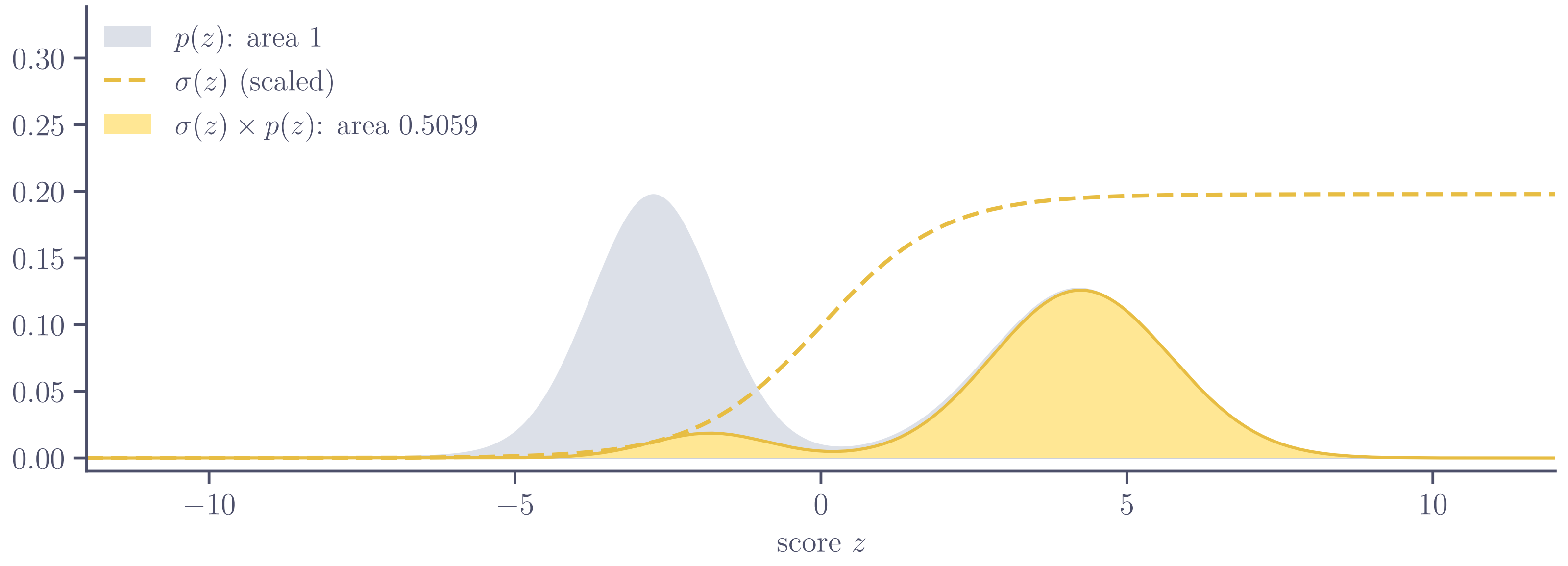
GOAL

Fit w on incomplete training points

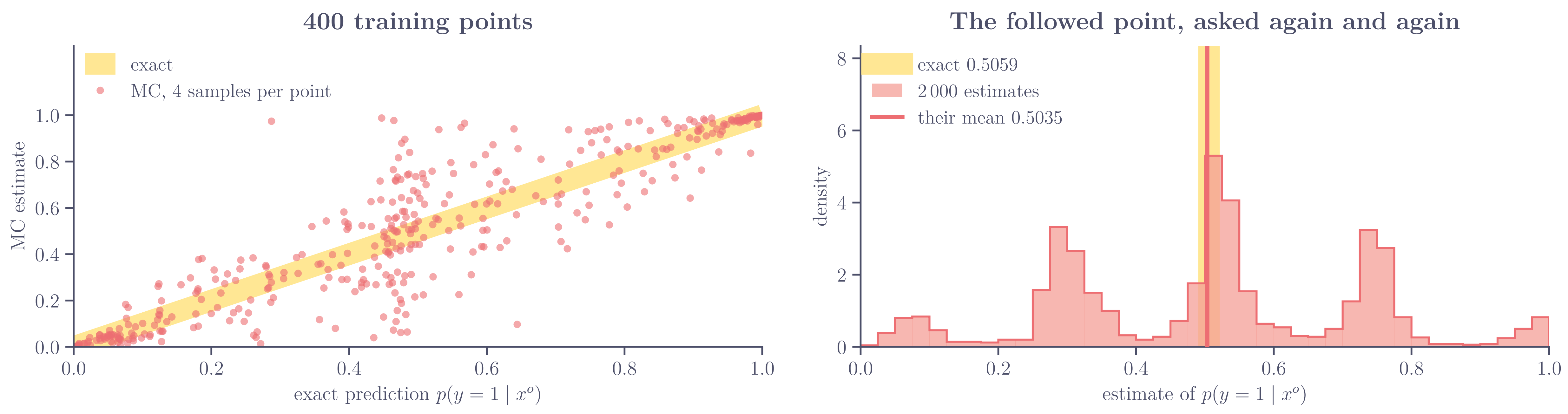
$$p(y=1 \mid x^o) = \mathbb{E}_{p(z|x^o)}[\sigma(z)] = \int \sigma(z)\, p(z)\, dz$$



In training, also the training points have missing features (here 60% of them, at random). The density model is fit first, its likelihood leaves the missing features out exactly. Then w is fit to the prediction that is made at test time: for a point with observed x^o it is the yellow area under $\sigma(z) \times p(z \mid x^o)$, here for a training point with 9 of 15 features missing and label $y=1$. Its loss is $-\log$ of the area of its label, and training minimises the mean loss over all points, $-\frac{1}{N}\sum_n \log p(y_n \mid x_n^o)$. So the area is needed for every point in every step, and the gradient of its log.

STEP 1

MC estimates every prediction from random fill-ins

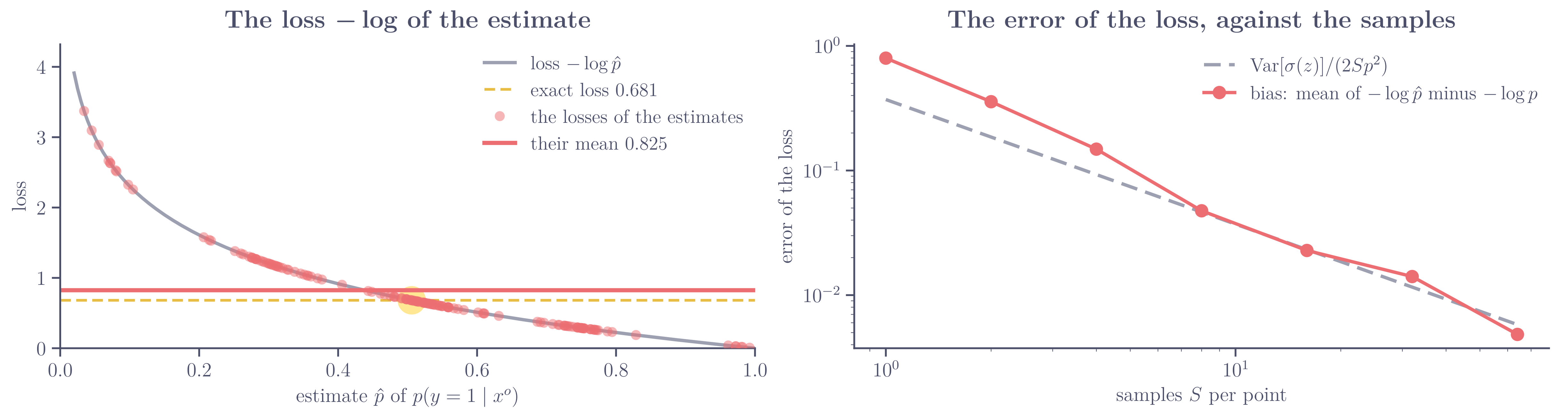

400 training points

The followed point, asked again and again

MC estimates the area for every training point by the average of $\sigma(z_i)$ over 4 fill-ins of the missing features, drawn from the model. Every dot is one of 400 training points: its MC estimate against the exact value. They scatter around the diagonal. The same point asked 2 000 times gives a different answer every time: a histogram around the exact value (yellow), its mean is right, single estimates are not.

STEP 2

The loss of a random estimate is too high on average

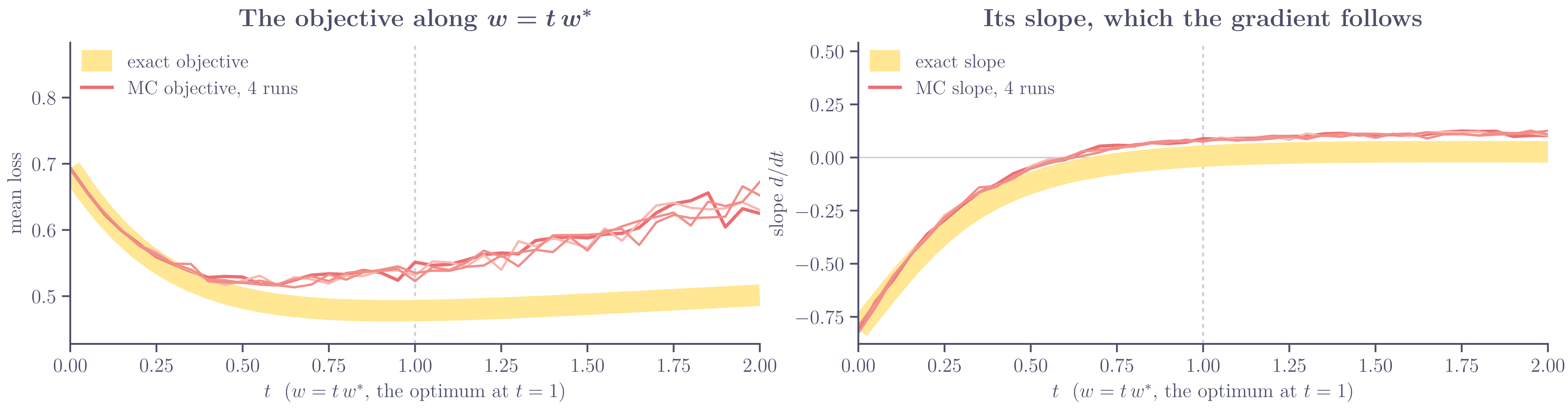

The loss $-\log$ of the estimate

The error of the loss, against the samples

The loss is $-\log$ of the prediction of the label. The curve bends upwards, so the losses of the random estimates (dots) are too high on average (Jensen): their mean 0.825 is above the exact loss 0.681. The bias of the loss, mean of $-\log\hat{p}$ minus $-\log p$, falls like $1/S$ with the number of samples S. It is largest where the prediction varies most over the missing features, so it punishes w for relying on them.

STEP 3

The objective is noisy and too high, its minimum moves

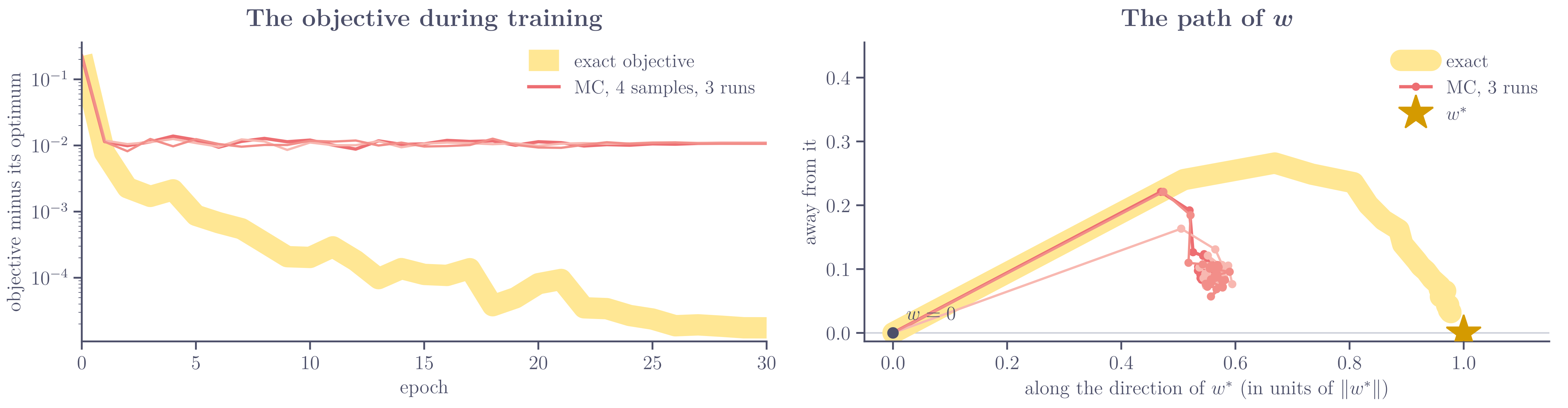

The objective along $w = t\,w^*$

Its slope, which the gradient follows

The mean loss over 2 000 training points along one direction of w, $w = t\,w^*$ (w^* the optimum of the exact objective, at $t=1$). Every MC run (new fill-ins at every t) is noisy and above the exact objective (yellow band), more so for larger w, so its minimum moves to a smaller w. Its slope is noisy and too high, so every step is pushed towards $w=0$: the bias acts like an extra penalty that shrinks w. With more samples it shrinks like $1/S$.

STEP 4

Training ends short of the optimum w^*


The objective during training

The path of w

Training with Adam on minibatches as in the experiments, new fill-ins in every step: the objective minus its optimum after every epoch, 3 runs. They stop well above the exact objective (yellow band), at the optimum of the biased MC objective. The path of w from $w=0$, along the direction of w^* and away from it (in units of $\|w^*\|$): every run is different, and all of them stop far short of w^* (star) and of the exact path (yellow), at a smaller w.